

VR Builds Lasting Empathy & Compassion: Observation Edges Embodiment in Adolescents

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ABSTRACT

We tested whether VR perspective-taking enhances empathy and compassion in adolescents, comparing first-person embodiment and third-person observation. In a randomized trial (N=703, ages 13-18, Kenya and Nigeria), participants experienced VR scenarios depicting ethnic discrimination in Embody (first-person), Observe (third-person), or Control conditions. Both VR conditions produced medium-to-large effects on empathy and compassion among low-baseline participants ($d=0.45-0.80$), with effects persisting at four-week follow-up. Observe outperformed Embody: in the full sample, Observe significantly increased empathy and compassion relative to Control at post-intervention ($d=0.21$) and follow-up ($d=0.30$). Path models revealed a suppression effect explaining Embody's underperformance: Embody increased social presence ($d=0.24$), which predicted higher compassion, but simultaneously exerted a direct negative effect on compassion ($c'=-3.84, p=.016$), partially cancelling the benefit. For emotionally intense content, third-person observation may cultivate empathy and compassion more effectively than first-person embodiment in adolescents.

Index Terms: Virtual reality, embodiment, children, adolescents, empathy, compassion, perspective-taking.

1 INTRODUCTION

Virtual Reality's promise for building empathy rests on the idea that users can "become someone else" and experience the world from another's perspective. Compassion adds a pro-social dimension to empathy, often the desire to do something to relieve a negative situation. First-person embodiment, where the user inhabits an avatar, sees through its eyes and has agency, is often assumed to maximize this perspective-taking potential, primarily because it is considered to cause a greater immersion, a deeper sense of presence or even body ownership. The result is a greater empathy, and the related property of compassion [1, 9].

We tested this assumption in adolescents (13-18 years) experiencing ethnic discrimination in Virtual Reality (VR) game scenarios. Our design compared two VR modes against a no-intervention control condition: Embody (first-person perspective) and Observe (third-person perspective). Both VR conditions involved similar discrimination experiences, only the visual perspective and agency differed. This study extends our prior work with Nigerian adolescents (authors, under review) to a larger, cross-national and cross-cultural sample spanning Nigeria and Kenya, and within each country multi and diverse cultures.

The results challenge conventional wisdom. VR worked and even showed persistence of effects. However, it did not work in the way embodiment theory predicts. While both the Observe and the Embody conditions produced significant and sustained gains in

empathy and compassion relative to Control, the results were more consistent for the Observe condition at all time points. The Observe condition outperformed the Embody condition. The Embody condition however produced higher social presence scores.

One theoretical reason for this counterintuitive finding is the burden of cognitive load imposed in the Embody condition during a stressful VR scenario, as well as the cognitive and affective distancing implied by the Observe condition which allows participants to achieve near perfect perspective taking. Path analysis however provides a more accurate explanation. The Embody condition operates through competing psychological pathways. On one hand, it increases social presence, which benefits the prosocial attribute of *Feeling for the Person Suffering*, a compassion subscale. On the other hand, it simultaneously exerts a direct negative effect on the same compassion measure, which partially cancels the earlier benefit. This suppression effect thus explains the null total effect of Embody relative to Observe and has practical implications for designing VR experiences for children.

Critically, the VR effects persisted to four-week follow-up. This addresses a significant gap in the literature, where the absence of longitudinal evidence has been repeatedly noted as a limitation undermining claims about VR's empathy-enhancing potential [16, 17]. Most prior studies have assessed effects only immediately post-intervention or at 1-2-week follow-up [9]. Our four-week follow-up with adolescents provides rare longitudinal evidence that VR-induced empathy gains can be sustained.

2 RELATED WORK

Embodiment in VR. Embodiment in VR refers to the first-person perspective that gives the user a sense of owning and controlling a virtual body [11]. Additionally, it confers agency and a sense of self-location within the virtual space. Visuo-proprioceptive integration is assumed to contribute to this perspective [7]. Most of the VR research to date has focused on adults with limited attention to whether these same mechanisms operate similarly in children. Bailey and Bailenson [2] note that children may experience VR as more real than adults, and that their still-developing sense of self may lower the threshold for responding to virtual content as if it were real.

Empathy versus Compassion. Empathy and compassion are related but distinct constructs. Empathy involves vicariously adopting another's emotion [4], which comprise both the cognitive understanding and affective sharing of another's experience [19]. Compassion adds to empathy the prosocial motivation to alleviate suffering [6, 15]. Perspective-taking can produce two distinct responses: empathic concern (other-focused, compassionate) and personal distress (self-focused, avoidance). When the content being experienced is emotionally intense, the balance may tilt towards personal distress rather than concern for another [3]. This possibility has not been systematically tested in children's VR.

Suppression Effects. Intervention effects can decompose into indirect and direct pathways that oppose and cancel out each other. This is called suppression [12]. When a beneficial indirect effect coexists with a harmful direct effect, total effects may appear null

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Table 1: Sample characteristics by condition

Characteristic	Control	Embodiment	Observe
N	247	224	232
Age M (SD)	16.60 (1.47)	16.67 (1.43)	16.50 (1.42)
Female n (%)	107 (43.32)	91 (40.62)	110 (47.41)
Kenya n (%)	132 (53.44)	122 (54.46)	121 (52.16)
Nigeria n (%)	115 (46.56)	102 (45.54)	111 (47.84)

even when meaningful processes operate. To detect suppression, there is a need for an explicit path decomposition.

3 METHODS

3.1 Participants

Participants were 703 adolescents (ages 13-18, 43.8% female) from four secondary schools, two in Kenya and two in Nigeria. The mean age was 16.59 (SD=1.44). All participants provided assent while parents/guardians provided consent. Ethics approval was granted by the first author's institution. Governments of both countries granted written research approval and permission. Table 1 presents sample characteristics by condition.

3.2 Design

The customized VR content was designed as a gamified experience depicting realistic scenarios of ethnic discrimination in a school setting. Part one took place inside a school bus on the way to school; the protagonist, as a member of a unique ethnic group, was verbally mocked by peers from other ethnic groups. Part two took place inside a classroom, where the protagonist faced social rejection along with insulting and discriminatory comments from classmates during a group activity. Avatars were realistic depictions that had similar physical characteristics as the study participants. Participants were individually randomised within schools to one of three conditions:

Embodiment (First-person perspective) (n=224): Participants embodied an avatar experiencing ethnic discrimination. Visual perspective was from inside the avatar's body; hand movements were tracked.

Observe (Third-person perspective) (n=232): Participants observed the same scenario but as passive bystanders. They witnessed discrimination happening to another character but did not embody the character, nor did their actions have any influence on game outcome or progression.

Control (n=247): No VR intervention. Participants completed assessments only, during the intervention session.

3.3 Measures

All outcomes used 0-100 visual analog scales captured via a digital tablet. **Embodiment** (15 items, $\alpha=0.94$; Peck & Gonzalez-Franco, 2021 [14]; sample adapted question "During the VR experience at some point it felt as if my real body was starting to take on the posture or shape of the avatar's body that I saw."): captured body ownership and agency in VR. **Social Presence** (6 items, $\alpha=0.80$; Herrera et al., 2018 [9]): sense of being with others in the virtual environment. Sample question: "How strongly did you feel that the human characters in the VR experience were present?". **Empathy** (AMES, 8 items, $\alpha=0.79$): Adolescent Measure of Empathy and Sympathy [18]. Empathy was captured with the composite measure and also with the subscales of cognitive and affective empathy. An alternative measure of Empathy used was the Fantasy subscale of the IRI (Interpersonal Reactivity Index, 5 items, $\alpha=0.81$) [5] which we adapted for the VR experience. One of the questions reads: "I imagine how I would feel if what is happening in a VR story happens to me", and another: "After playing a VR game, I feel like I am one of the characters". **Compassion** (SOCS,

Table 2: Selected main effects (full sample)

Outcome	Time	Contrast	d	p	P _{idr}
IRI Fantasy	T2	Embodiment	0.15	.081	.122
IRI Fantasy	T2	Observe	0.21	.011	.032*
IRI Fantasy	T3	Embodiment	0.14	.172	.172
IRI Fantasy	T3	Observe	0.30	.004	.012*
Sympathy	T3	Observe	0.36	<.001	.002*
Sympathy	T3	Emb-Observe	0.27	.012	.017*

*P_{idr} < .05

20 items, $\alpha=0.85$): Sussex-Oxford Compassion Scale [8]. This was measured as a composite as well as each of its five subscales. Outcomes were assessed at baseline (T1), post-intervention (T2) and four-weeks follow-up (T3).

3.4 Analysis

We used mixed-effects ANCOVA models with baseline scores as covariates, controlling for age, gender, and school (as random effect). Effect sizes (Cohen's d) were computed using pooled baseline standard deviations. Benjamini-Hochberg correction addressed multiple comparisons. We then used path models to decompose the Embodiment vs. Observe comparison into indirect effects (through social presence) and direct effects. We used multiple imputation (m=20) to address differential attrition.

4 RESULTS

4.1 Main Effects: Full Sample

Mixed-effects models revealed limited main effects in the full sample. After FDR correction, significant effects emerged for IRI-Fantasy (Empathy measure): at T2, Observe significantly outperformed Control (d=0.21, p=.032). At T3, the Observe effect persisted and strengthened (d=0.30, p=.012). No significant main effects were observed for AMES or SOCS in the full sample. Table 2 presents selected main effects.

4.2 Manipulation Check

The Embodiment condition produced significantly higher social presence than Observe (M=76.37 vs. 71.84; t(414)=2.46, p=.014, d=0.24). Contrary to expectations, measured embodiment did not differ between conditions (M=77.37 vs. 74.40; t(412)=1.55, p=.122, d=0.15). This pattern suggests that the perspective manipulation affected how participants related to other characters in the virtual environment (social presence), rather than to their own virtual body (embodiment). Adolescents in both Embodiment and Observe conditions may have experienced avatar identification through narrative engagement. On the other hand, the first-person condition uniquely intensified awareness of the discriminatory characters, triggering self-referential processing that contributed to the suppression effect on compassion outcomes.

4.3 Baseline-Stratified Effects

Among participants in the lowest baseline tertile, both VR conditions produced substantial improvements at T2, with medium-to-large effect sizes (see Table 3). At four-week follow-up (T3), effects attenuated considerably. Sustained significant effects were observed for IRI-Fantasy (Embodiment: d=0.39, p=.038; Observe: d=0.61, p=.007).

4.4 Suppression Effect

Path models revealed competing pathways from the Embodiment condition to compassion outcomes (see Table 4). Path a (Embodiment to Social Presence): b=4.64, SE=1.84, p=.012. Path b (Social Presence to Compassion measure: Feeling for the Person Suffering): b=0.43, SE=0.04, p<.001. Path c (Direct effect): b=-3.84,

Table 3: Low baseline subgroup effects at T2

Outcome	Contrast	d	p_{idr}	Sig
AMES Composite	Embody	0.62	<.001	*
AMES Composite	Observe	0.77	<.001	*
SOCS Composite	Embody	0.45	.022	*
SOCS Composite	Observe	0.61	.001	*
SOCS Feeling	Observe	0.79	<.001	*
IRI Fantasy	Embody	0.61	<.001	*
IRI Fantasy	Observe	0.80	<.001	*

Table 4: Suppression effect path analysis

Path	Estimate	SE	p
a: First-person to Social Presence	4.64	1.84	.012
b: Social Presence to SOCS Feeling	0.43	0.04	<.001
c: Direct effect	-3.84	1.59	.016
Indirect effect (a $\times$ b)	1.98	0.81	.014
Total effect (c + a $\times$ b)	-1.86	1.76	.291

Model fit: $\chi^2(2) = 11.56$, $p = .003$. $N = 414$.

SE=1.59, $p=.016$. Indirect effect (a $\times$ b): $b=1.98$, SE=0.81, $p=.014$. Total effect: $b=-1.86$, SE=1.76, $p=.291$ (n.s.). This shows suppression where Embody (first-person VR) simultaneously helps (via Presence) and hurts (via the direct negative pathway) compassion.

4.5 Comparing First Person vs Third Person

At T3, Observe significantly outperformed Embody on SOCS Universality of Suffering compassion measure ($d=-0.23$, $p=.038$) and IRI Fantasy, empathy measure ($d=-0.22$, $p=.046$). Despite Embody resulting in greater social presence, Observe produced larger and more sustained effects.

5 DISCUSSION

The central finding is that VR perspective taking leads to increased empathy and compassion in adolescents, but the effect strength and breadth depends on the perspective adopted in the VR experience. Theory suggests that the Embody first person perspective which has more agency, body ownership and sense of place should result in a greater immersion and hence greater perspective taking, than the passive, third person Observe. While both modes increased empathy and compassion, with effects persisting to follow-up, the Observe mode produced larger and more consistent gains relative to Control. Rather than add value over third person bystander observation, the Embody mode undermined outcomes through a direct negative pathway.

The suppression pattern provides us with a mechanistic explanation for the counterintuitive finding about the Embody condition. While the first-person perspective may increase Social Presence, it also decreased the Compassion measure, Feeling for the Person Suffering. It would appear that being immersed in VR in the first person may trigger self-referential distress which may interfere with other-focused compassion. A VR experience of an emotionally intense situation may also produce an affective overload which causes distress in the participant. The third person Observe mode may provide the psychological distance that protects against this emotional and cognitive overload, allowing for a more dispassionate processing of the situation, hence leading to increased perspective taking, empathy and compassion. This interpretation aligns with Iriye and St. Jacques [10] who found that third-person VR experiences reduced subjective emotional recollection compared to first-person experiences. It also aligns with our prior work with Nigerian adolescents, which found the Observe mode to be superior to the Embody mode.

5.1 Persistence of Effects

A notable finding is that both VR conditions produced effects that persisted to four-week follow-up. In the low-baseline group, significant effects remained for IRI-Fantasy in both Embody ($d=0.39$, $p=.038$) and Observe ($d=0.61$, $p=.007$) conditions. This addresses a critical gap in the VR empathy literature. Reviews have consistently noted the “absence of longitudinal studies” and “little knowledge about the medium and long-term effect of the interventions” [16]. A recent systematic review concluded that “no longitudinal studies have been conducted to date” to establish whether VR interventions produce lasting changes (Journal of New Approaches in Educational Research 2025). Our findings provide evidence that VR perspective-taking can produce durable empathy gains in adolescents, not merely transient effects attributable to novelty.

5.2 Implications for VR Design

These findings have implications for designing immersive experiences to ensure and promote psychological safety in children’s VR embodiment experiences. For emotionally intense content, designers should be cautious about first-person perspectives, considering them as likely unsafe and generally suboptimal. Third-person (Observe) perspectives may provide equal or even greater engagement for perspective-taking than the first-person mode, while avoiding the negative direct effect observed in our study.

5.3 Limitations

Differential attrition threatens internal validity. The higher dropout in the Observe condition (35.8% vs. 17.4% for Embody) raises questions about whether participants are representative. We addressed this with multiple imputations, but the concern remains. The negative direct effect, while significant, was modest in magnitude. We inferred but did not measure personal distress directly, which limits mechanistic conclusions.

6 CONCLUSION

Playing VR games depicting ethnic discrimination can enhance empathy and compassion in adolescents, particularly those with lower baseline scores. Both the first-person and third-person VR modes produced immediate effects that persisted to four-week follow-up, providing longitudinal evidence that has been notably absent from the VR empathy literature. Third-person observation yielded larger and more consistent benefits. For emotionally intense content depicting ethnic discrimination, first-person immersion in VR may attenuate compassion through a direct negative pathway that partially cancels the benefits of enhanced social presence. Designers of VR experiences for children should consider whether perspective-taking goals are better served by observation than by embodiment.

ACKNOWLEDGMENTS

This work was supported by Meta.

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